

The Multi-Player Stochastic Commitment-Race Model over the Finite Standard Nuclear oliGARCHy Architecture

A Contribution to the Standard Nuclear oliGARCHy

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Abstract

This paper develops a multi-player stochastic commitment-race model over the finite Standard Nuclear oliGARCHy (SNoG) architecture. The central strategic claim is that, conditional on the finite structural architecture and the inevitability result constraining the admissible terminal configurations, strategic competition need not be represented primarily as an arms race. Instead, rational players may compete over the timing and credibility of commitments to structurally admissible positions.

Each player possesses an ambiguity option: delaying commitment preserves flexibility and information value. Delay, however, is costly because other players may commit, coordinate, alter expectations, or otherwise change the continuation game. The resulting problem is formulated as a stochastic dynamic game combining optimal control, optimal stopping, strategic interaction, and finite-state structural constraints.

The model introduces commitment times, stochastic arrival hazards, commitment payoffs, transition costs, and equilibrium conditions. A late-entry player may reason backward from a constrained terminal architecture rather than attempting to discover the architecture through forward experimentation. When multiple players possess the same structural knowledge, however, backward reasoning itself generates strategic interaction. The result is a commitment race: players must balance the option value of waiting against the risk that delay permits other players to change the equilibrium.

The paper argues that the relevant strategic resource is therefore not necessarily incremental capability accumulation, but the ability to make a credible commitment to a structurally admissible position at an appropriate stochastic stopping time.

The paper ends with “The End”

1 Introduction

Consider a strategic environment containing a finite set of structurally admissible positions. Let the SNoG architecture be represented by

$$\mathcal{D} = \{d_1, d_2, \dots, d_9\},$$

where the elements of $\mathcal{D}$ are the nine districts of the finite SNoG architecture.

Suppose further that the structural theory constrains the set of admissible terminal configurations. Then the strategic problem facing a player is not necessarily one of discovering an unconstrained future. Instead, the player may reason backward from a restricted set of possible structural outcomes.

This changes the nature of competition.

The conventional arms-race representation emphasizes:

capability accumulation $\longrightarrow$ relative advantage.

The present framework instead emphasizes:

structural knowledge $\longrightarrow$ backward reasoning $\longrightarrow$ commitment timing $\longrightarrow$ equilibrium selection.
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The strategic problem is therefore a commitment race.

A player that delays commitment retains flexibility. Yet delay also permits other players to alter the strategic environment. Consequently, waiting is not a neutral action.

Let τ_i denote the commitment time of player i . Then the strategic ordering

$$\tau_{(1)} \leq \tau_{(2)} \leq \dots \leq \tau_{(n)}$$

may itself influence the terminal configuration.

The central question is therefore:

When should a rational player commit to a structurally admissible position when other players are solving the same timing problem?

2 The Finite SNoG Architecture

Let

$$\mathcal{D} = \{d_1, d_2, \dots, d_9\}$$

denote the finite set of SNoG districts.

Let $\mathcal{S}$ denote the set of structurally admissible configurations:

$$\mathcal{S} \subseteq 2^{\mathcal{D}}.$$

The terminal state of the system is therefore constrained:

$$X_T \in \mathcal{S}.$$

The significance of finiteness is substantial. A player need not solve an unbounded search problem. Instead, the player faces a constrained strategic space.

Define the terminal architecture as

$$\mathcal{S}^* \in \mathcal{S}.$$

A late-entry player can therefore formulate the problem in reverse:

$$\boxed{\mathcal{S}^* \longrightarrow \text{admissible predecessor states} \longrightarrow \text{optimal transition path.}}$$

This is the informational advantage of late entry.

However, if several players recognize the same structural constraints, each may attempt to optimize its own path toward an admissible terminal state.

Thus:

$$\text{common structural knowledge} \not\Rightarrow \text{absence of strategic competition.}$$

Instead:

$$\boxed{\text{common structural knowledge} \Rightarrow \text{competition over commitment.}}$$

3 Players, States, and Controls

Let

$$\mathcal{I} = \{1, 2, \dots, n\}$$

denote the set of strategic players.

For each player i , define the state vector

$$X_t = (Q_t, C_t, B_t, R_t, Z_t),$$

where:

Q_t : the structural configuration of the finite SNoG system;

C_t : the vector of public commitment states;

B_t : the vector of beliefs held by strategic players;

R_t : the vector of strategic responses and relative positions;

Z_t : the exogenous stochastic environment.

Player i chooses a control process

$$u_i(t) \in \mathcal{U}_i$$

and a stopping time

$$\tau_i.$$

The control may represent non-terminal strategic choices, while τ_i represents the time at which player i makes a relevant credible commitment.

The joint control profile is

$$u(t) = (u_1(t), u_2(t), \dots, u_n(t)).$$

4 Stochastic State Dynamics

Let the strategic state evolve according to

$$dX_t = b(X_t, u_t) dt + \Sigma(X_t, u_t) dW_t + J(X_t, u_t) dN_t,$$

where:

b : is the deterministic drift;

Σ : is the diffusion matrix;

W_t : is a vector Brownian motion;

N_t : is a jump process representing discrete strategic events;

J : maps strategic jumps into changes in the state.

The jump component is important because commitments by other players can change the strategic environment discontinuously.

For example,

$$X_{\tau_j^-} \neq X_{\tau_j^+}$$

when player j makes a commitment that changes the beliefs or feasible strategies of other players.

Thus:

another player's commitment = an endogenous stochastic shock.

5 The Value of Ambiguity

Before commitment, player i possesses an ambiguity option.

Let

$$V_i^A(X_t)$$

denote the continuation value of remaining uncommitted.

Let

$$V_i^C(X_t)$$

denote the value of commitment.
The commitment gain is

$$\Delta_i(X_t) = V_i^C(X_t) - V_i^A(X_t).$$

A simple individual stopping rule would be

$$\tau_i^* = \inf \{t \geq 0 : \Delta_i(X_t) \geq 0\}.$$

But this rule is incomplete in a strategic environment because Δ_i depends on the expected commitment behavior of other players.

Therefore:

$$\Delta_i = \Delta_i(X_t, \mathbb{E}_t[\tau_{-i}]).$$

The optimal stopping boundary is endogenous.

6 The Multi-Player Commitment Race

Let

$$\tau = (\tau_1, \tau_2, \dots, \tau_n)$$

be the vector of commitment times.
Player i solves

$$V_i(x) = \sup_{u_i, \tau_i} \mathbb{E}_x \left[\int_0^{\tau_i} e^{-\rho t} \Pi_i^A(X_t, u_t) dt + e^{-\rho \tau_i} K_i(X_{\tau_i}, \tau_i, \tau_{-i}) + \int_{\tau_i}^T e^{-\rho t} \Pi_i^C(X_t, u_t) dt + e^{-\rho T} G_i(X_T) \right].$$

Here:

Π_i^A : is the payoff while ambiguity is maintained;

K_i : is the commitment payoff, which depends on the timing of other players;

Π_i^C : is the payoff after commitment;

G_i : is the terminal structural payoff.

The strategic feature enters through:

$$K_i = K_i(X_{\tau_i}, \tau_i, \tau_{-i}).$$

Thus the value of committing now depends on whether another player committed earlier, simultaneously, or later.

7 Commitment Hazards

Let

$$\lambda_j(X_t)$$

denote the conditional hazard that player j makes a strategically relevant commitment.
The aggregate competitive arrival hazard facing player i is

$$\Lambda_{-i}(X_t) = \sum_{j \neq i} \lambda_j(X_t).$$

If delay exposes player i to the risk that another player changes the game, then the value of waiting is discounted not only by time preference ρ but also by strategic arrival risk.

A reduced representation is:

$$V_i^A = \mathbb{E} \left[\int_0^{\tau_i} e^{-\int_0^t (\rho + \Lambda_{-i}(X_s)) ds} \Pi_i^A(X_t) dt + \dots \right].$$

Hence:

$$\Lambda_{-i} \uparrow \Rightarrow \text{the strategic cost of waiting may increase.}$$

This is the fundamental mathematical distinction between ordinary ambiguity and ambiguity inside a commitment race.

8 Backward Reasoning by Late-Entry Players

Suppose the terminal structural configuration is constrained by

$$X_T \in \mathcal{S}^*.$$

A late-entry player can solve:

$$\min_{u_i, \tau_i} \mathbb{E}[\mathcal{C}_i]$$

subject to

$$X_T \in \mathcal{S}^*.$$

The player does not need to discover every possible terminal architecture. Instead:

$$\text{terminal constraint} \rightarrow \text{feasible predecessor set} \rightarrow \text{optimal present action.}$$

Let

$$\mathcal{P}_k(\mathcal{S}^*)$$

denote the set of states that can reach the terminal architecture in k steps. Then:

$$\mathcal{P}_{k+1}(\mathcal{S}^*) = \{x : \exists u \text{ such that } F(x, u) \in \mathcal{P}_k(\mathcal{S}^*)\}.$$

Backward construction therefore generates the feasible strategic region.

9 The Strategic Paradox of Shared Backward Reasoning

Suppose every player understands the same terminal architecture.

Then each player may reason:

$$\mathcal{S}^* \rightarrow \text{required future position} \rightarrow \text{optimal present commitment.}$$

But because every player's action enters the transition function,

$$X_{t+1} = F(X_t, u_1(t), \dots, u_n(t), \varepsilon_{t+1}),$$

one player's backward solution changes another player's problem. Therefore:

$$\text{the solution path is itself strategically endogenous.}$$

This produces a recursive commitment race.

10 Commitment-Race Equilibrium

A commitment strategy for player i is a pair

$$\sigma_i = (u_i, \tau_i).$$

Let

$$\sigma = (\sigma_1, \dots, \sigma_n).$$

A stochastic commitment-race equilibrium is a profile

$$\sigma^*$$

such that for every player i ,

$$V_i(\sigma_i^*, \sigma_{-i}^*) \geq V_i(\sigma_i, \sigma_{-i}^*)$$

for all admissible alternative strategies σ_i .

Equilibrium therefore requires both:

control optimality

and

stopping-time optimality.

11 A Variational Characterization

Let $\mathcal{L}^u$ denote the infinitesimal generator of the controlled stochastic process.

The value function satisfies a strategic variational inequality of the form

$$\max \left\{ \sup_{u_i} \left[\mathcal{L}^{u_i, u_{-i}^*} V_i - \rho V_i + \Pi_i^A \right], V_i^C - V_i \right\} = 0.$$

The first term governs continuation under ambiguity.

The second term governs stopping and commitment.

Therefore the state space divides into:

$$\mathcal{A}_i = \{x : V_i^A(x) > V_i^C(x)\},$$

the ambiguity region, and

$$\mathcal{C}_i = \{x : V_i^C(x) \geq V_i^A(x)\},$$

the commitment region.

The free boundary is

$$\mathcal{B}_i = \{x : V_i^A(x) = V_i^C(x)\}.$$

Because other players influence both value functions:

$$\mathcal{B}_i = \mathcal{B}_i(\sigma_{-i}^*).$$

Thus the commitment boundary is a strategic object.

12 The Commitment Cascade

Suppose player j commits at time τ_j .

This may alter player i 's commitment gain:

$$\Delta_i(X_{\tau_j^+}) - \Delta_i(X_{\tau_j^-}) \neq 0.$$

If

$$\Delta_i(X_{\tau_j^-}) < 0$$

but

$$\Delta_i(X_{\tau_j^+}) \geq 0,$$

then player j 's commitment triggers player i 's commitment.

This creates a cascade:

$$\tau_j \rightarrow \Delta_i \uparrow \rightarrow \tau_i \rightarrow \Delta_k \uparrow \rightarrow \tau_k.$$

Hence a commitment race can generate endogenous phase transitions:

isolated waiting $\longrightarrow$ first commitment $\longrightarrow$ belief revision $\longrightarrow$ commitment cascade.

13 No Necessary Monotonicity of Early Commitment

The first mover does not necessarily obtain the highest payoff.

Let

$$F_i = V_i(\tau_i < \tau_{-i})$$

be the first-mover value and

$$S_i = V_i(\tau_i > \tau_{-i})$$

the follower value.

Three broad cases are possible:

$$F_i > S_i,$$

in which first commitment is advantageous;

$$F_i < S_i,$$

in which waiting for another player is advantageous;

and

$$F_i = S_i,$$

in which commitment order is strategically neutral.

Therefore the commitment race is not simply:

$$\text{commit earlier} = \text{win}.$$

Instead, the strategic problem is:

commit at the optimal point before delay changes the equilibrium adversely.

14 The Cost of Delay

Let the value of waiting be

$$W_i(t).$$

Then its time derivative can be decomposed as:

$$\frac{dW_i}{dt} = \underbrace{\frac{\partial W_i}{\partial t}}_{\text{ordinary time effect}} + \underbrace{\frac{\partial W_i}{\partial X} \frac{dX}{dt}}_{\text{state evolution}} - \underbrace{\Lambda_{-i} L_i}_{\text{competitive commitment risk}},$$

where L_i is the expected loss associated with another player changing the continuation game. The critical condition for commitment is therefore not simply:

$$V_i^C > V_i^A.$$

It may instead be represented as:

$$V_i^C > V_i^A + \text{expected future option value} - \text{expected competitive delay loss}.$$

As competitive hazards increase, the optimal stopping boundary may move.

15 The Structural Scarcity of Commitment

In a finite architecture, strategic positions may be structurally scarce.

Let

$$\mathcal{O} = \{o_1, \dots, o_m\}$$

denote the set of strategically relevant commitment positions.

If

$$m < n,$$

then not every player can occupy every position simultaneously.

The allocation problem becomes:

$$a : \mathcal{I} \rightarrow \mathcal{O}.$$

If a commitment changes the feasible allocation set,

$$\mathcal{A}_{t+} \subsetneq \mathcal{A}_{t-},$$

then commitment has an exclusion effect.

This provides a formal mechanism through which delay may become costly even without additional capability accumulation.

16 Arms Race versus Commitment Race

The distinction between the two strategic systems can be summarized as follows.

Feature	Arms Race	Commitment Race
Primary strategic variable	Capability	Credible commitment
Main action	Accumulation	Timing and signaling
Scarce resource	Material capability	Structurally credible position
Main uncertainty	Relative capability	Future commitments
Dynamic mechanism	Competitive accumulation	Optimal stopping interaction
Critical risk	Capability gap	Adverse equilibrium change
Late-entry advantage	Learning	Backward reasoning

The present model does not claim that material capabilities are irrelevant.

Rather, it proposes that once a finite structural architecture constrains the strategically relevant destination, the marginal strategic problem may shift from capability acquisition toward equilibrium selection and commitment timing.

17 A Commitment-Race Theorem

Theorem 1 (Commitment-Race Principle). *Suppose that:*

1. *the terminal strategic architecture belongs to a finite admissible set $\mathcal{S}$;*
2. *each player can infer a nonempty set of feasible paths toward $\mathcal{S}^*$;*
3. *each player receives a positive option value from delaying commitment;*
4. *the commitment of any player can alter the continuation value of at least one other player;*
5. *the hazard of competing commitment is strictly positive on a nonempty subset of the state space.*

Then the optimal commitment problem of each player is strategically interdependent. In general, the optimal stopping time cannot be determined from the player's individual state alone.

Proof. By assumption, player i receives a continuation value from delay.

However, by assumptions (4) and (5), the continuation process is affected by random commitment times of other players.

Therefore the state transition law relevant to player i depends on

$$\tau_{-i}.$$

Consequently:

$$V_i = V_i(X_t, \tau_i, \tau_{-i}).$$

Thus the optimal stopping time satisfies

$$\tau_i^* = \Gamma_i(X_t, \sigma_{-i}^*),$$

rather than depending on X_t alone.

Hence commitment timing is strategically interdependent, producing a multi-player commitment race. $\square$

18 Implications for a Late-Entry Player

A late-entry player has two simultaneous advantages and disadvantages.

The advantage is:

observed structural history + finite terminal architecture $\rightarrow$ backward reasoning.

The disadvantage is:

other late or catch-up players may reason backward as well.

Therefore:

information advantage $\nrightarrow$ strategic monopoly.

The player must solve:

$$\min_{\tau_i, u_i} \mathbb{E} [\text{transition cost} + \text{delay cost}]$$

subject to:

$$X_T \in \mathcal{S}^*,$$

while anticipating:

$$\tau_j^* = \Gamma_j (X_t, \sigma_{-j}^*)$$

for every other player j .

19 Conclusion

The finite SNoG architecture permits a different representation of strategic competition.

The relevant dynamic need not be modeled exclusively as an arms race:

more capability $\rightarrow$ greater advantage.

Instead, when the structural destination is constrained and strategically intelligible, rational players may reason backward from the admissible architecture.

If several players do so simultaneously, the result is a stochastic commitment race:

finite architecture	$\rightarrow$	backward reasoning	$\rightarrow$	strategic stopping	$\rightarrow$	commitment competition	$\rightarrow$	equilibrium selection
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The central decision variable is the commitment time:

$$\tau_i.$$

Waiting preserves option value:

$$V_i^A.$$

Commitment may stabilize a player's structural position:

$$V_i^C.$$

But competing commitments create an endogenous cost of delay:

$$\Lambda_{-i} L_i.$$

The optimal strategy is therefore neither perpetual ambiguity nor necessarily immediate commitment. It is the solution to:

$\tau_i^* = \arg \max_{\tau_i} \mathbb{E} [V_i (X_{\tau_i}, \tau_i, \tau_{-i}^*)]$
--

The resulting framework transforms the strategic interpretation of a finite SNoG architecture.

The fundamental race is not necessarily:

Who can build more?

It may instead be:

Who can identify the structural destination, reason backward from it, and commit at the optimal time before the commitments of others change the game?

That is the multi-player stochastic commitment-race problem over the finite SNoG architecture.

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